high wattage rating deliver high volume of sound and also it avoids the burn-out of voice coil and torn-out of cone paper. Manufactures determine the wattage rating of amplifiers and speakers. It is noted in the label pasted in the speakers. The wattage rating of speakers used in radio and TV receivers is from 0.5 watts to 15 watts approximately.

2.4.4 Column Loudspeakers

Many Dynamic Loudspeakers (in same wattage rating) used in single system are termed as Column Loudspeakers. They are placed vertically and closely to each other in a box. This system is placed in such a way that the most of the sound delivered from this box is towards audience. Since only a little amount of sound goes towards the wall and ceiling, echo is not produced. This system is used in auditorium and theaters. Figure 2.28 shows a Column loudspeaker.



Figure 2.28 Column Loudspeaker

2.4.5 Horn type Loudspeaker

This type of speakers is extensively used in Public meeting. It is designed in tube type funnel. It is more suitable than cone paper type speakers for delivering very high volume. It has very good frequency response in medium and high frequency audio range and less response in low frequency. A horn type loudspeaker is shown in Figure 2.29.



Figure 2.29 Horn Type Loudspeaker

2.4.6 Cross Over Network of Speakers

The method of connection needed when a woofer and tweeter are used in the audio system is called cross over network, since tweeters perform very well only in high frequencies and not in low frequencies. A simple cross over network with a capacitor connected in series with tweeter is shown in Figure 2.30. If the audio frequency decreases, the capacitive reactance of the capacitor increases. In a particular frequency, the capacitive reactance and the impedance of the tweeter will become equal. That frequency is referred as the 'cross over frequency' of the system.

A filter circuit is used to give the respective audio range of frequencies to the woofer and tweeter separately. This filter circuit has a coil and capacitor. Capacitor blocks the low frequency audio signal whereas coil prevents the entry of high frequency audio signal to the tweeter and woofer, respectively.

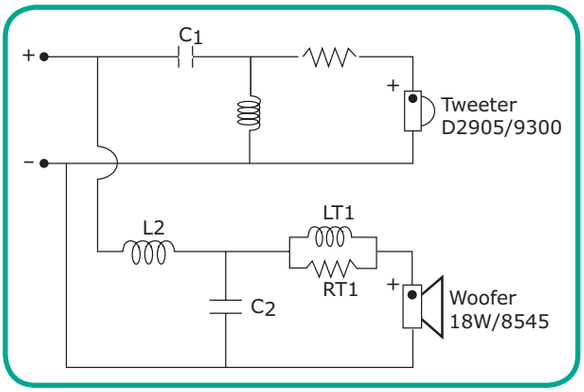


Figure 2.30 Cross Over Networks of Speakers

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